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Growth and chemical responses of balsam fir saplings released from intense browsing pressure in the boreal forests of western Newfoundland, Canada

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ABSTRACT

For ungulates, balsam fir (*Abies balsamea* (L.) Mill.) is thought to be a low quality forage due to high levels of indigestible fiber and secondary metabolites. Regardless, intensive browsing of fir saplings by overabundant moose (*Alces alces* L.) populations has led to the failure of fir-dominated forests in Newfoundland, Canada to regenerate following disturbance. The construction of moose exclosures in Gros Morne National Park (GMNP), western Newfoundland, presented an opportunity to compare levels of secondary metabolites (e.g. phenols and tannins), thought to act as anti-herbivore allelochemicals in balsam fir, under extremes of moose browsing pressure. We examined browsing damage to fir saplings and compared levels of carbon (C), nitrogen (N), total phenols and condensed tannins in current-year foliage outside and inside of eight moose exclosures. Outside of exclosures, 74% of fir saplings showed apical browsing while 97% were browsed laterally. At five exclosures, exposed saplings had significantly greater total phenol concentrations than inside exclosures. Such trends for condensed tannins were observed at only two exclosures, both locations having relatively high moose density and browsing damage. Unlike condensed tannins, total phenols were responsive to different patterns and levels of browsing. Foliar phenol concentration was negatively related to canopy closure, and to height and diameter of saplings, and positively related to the foliar C level. Foliar tannin level was unrelated to canopy or growth characteristics but positively related to the foliar N concentration. Moose browsing affected the defense chemistry of balsam fir; however, phenols and tannins responded differently, this likely being influenced by time of year. Foliar nitrogen levels were greater for inside saplings at only the two exclosures having highest moose density and browsing damage. Our data suggest that balsam fir employs constitutive chemical defense; however, this investment appears to be ineffective in deterring moose browsing.

1. Introduction

In Newfoundland, Canada, where moose (*Alces alces* L.) were introduced as recently as 1904 (Pimlott, 1953), there were between 125,000 and 150,000 moose on the island by 2003 (McLaren et al., 2004). A lack of predators and competitors are significant factors that contributed to the rapid growth of moose populations. Here, the last native wolf was shot in 1911 (Bergerud, 1971) and the only other ungulate, the native caribou (*Rangifer tarandus* L.), has little niche overlap with moose (Seip, 1992; Mahoney and Virgl, 2003). Due to a lack of hunting, impacts of moose overabundance became especially acute in Newfoundland's National Parks (Gosse et al., 2011). The forests of Gros Morne National Park (GMNP) in western Newfoundland have supported the densest population of moose in the world, with a reported crude density reaching 5.9 moose km^{-2} and ecological densities exceeding 18 moose km^{-2} in three different sampling areas (Taylor and Knight, 2009). Of the forested area in GMNP disturbed in recent decades, usually following outbreaks of spruce budworm (*Choristoneura fumiferana* Clem) or hemlock looper [*Lambdina fiscellaria fiscellaria* Guenée]), regeneration of balsam fir (*Abies balsamea* (L.) Mill.) has failed or been greatly impaired on 74% of these landscapes due specifically to intense winter browsing of saplings by moose (Taylor and Sharma, 2010). Prior to the time of high moose density, natural regeneration of insect-disturbed forests was consistently successful.

In GMNP, the proportion of balsam fir in the winter diet of moose is thought to exceed 90%; however, fir represents only 1% to 17% of the winter diet across much of the eastern and central North American range of moose (Joyal, 1976; Peek et al., 1976; McNicol and Gilbert, 1980; Thompson and Vukelich, 1981; Raymond et al., 1996; Routledge and Roese, 2004). The strict dependence of moose on a single conifer

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species appears to defy several commonly held principles related to tree-browser interactions and raises the question of why moose in Newfoundland eat so much balsam fir. Mammalian herbivores tend to avoid conifer species because of high levels of indigestible fiber and plant secondary metabolites (PSM) (Byrant et al., 1991; Burney and Jacobs, 2011). Like many conifers, balsam fir produces various classes of PSM including simple phenols, polyphenols (e.g. condensed tannins) and terpenes; presumably as a defense against herbivores (Thompson et al., 1989; Bauce et al., 1994; Dumont et al., 2005; Carlow et al., 2006; Sauvé and Côté, 2007; Grégoire et al., 2014). A diet of only one or two plant species is thought to be insufficient to maintain a healthy moose population (Peterson, 1978). Mammalian herbivores tend to have diverse diets to avoid the high energy expenditure required to detoxify a nutritious but chemically defended plant (Dearing et al., 2000; Parikh et al., 2017). Balsam fir has been described by some authors as having low nutritional and energetic quality with respect to ungulate diets (Ullrey et al., 1968; Crête and Courtois, 1997; Lefort et al., 2007); however, fir is relatively palatable compared to other dominant boreal conifers (e.g. white spruce (*Picea glauca* (Moench) Voss), black spruce (*P. mariana* (Mill.) BSP) (Thompson et al., 1992; Potvin et al., 2003; Dumont et al., 2005; Sauvé and Côté, 2007).

Given that balsam fir is intensively consumed by various insect and mammalian herbivores, it is surprising that so little is known about chemical defense mechanisms in this tree species. Phenolic compounds (e.g. total phenols and condensed tannins) are commonly examined in studies of tree-herbivore interactions as these PSM have been shown to act as allelochemicals that deter herbivores through toxicity and/or reduction of forage quality (Bryant et al., 1991; Spalinger et al., 2010; Barbehenn and Constabel, 2011; Grégoire et al., 2014). Under herbivory pressure, these allelochemicals are either maintained in high concentrations (constitutive defenses) or are produced following herbivore disturbance (induced defenses) (Karban and Myers, 1989; Herms and Mattson, 1992; Bixenmann et al., 2016). Several studies have examined the concentration of allelochemicals in balsam fir in relation to feeding behavior and development of herbivores (Thompson et al., 1989; Mattson et al., 1991; Sauvé and Côté, 2007) and the effect of environmental and forest management factors on tree resistance to insects (Fuentealba and Bauce, 2012, 2016); however, chemical defense mechanisms in direct response to herbivory have not been studied with the exception of a greenhouse study of fir seedlings (Nosko and Embury, 2018). The high snowfall and deep snowpack that characterizes western Newfoundland means that balsam fir seedlings tend not to be detected at the time that moose have shifted to their winter diet. As with other tree species, intense browsing pressure on balsam fir saplings should favor constitutive over inducible defenses (Bixenmann et al., 2016). Furthermore, a study of the nutritional quality of host trees used by spruce budworm concluded that there is little evidence of short-term induction of chemical defenses in balsam fir foliage (Mattson et al., 1991). In response to herbivory, resource allocation in plants can be directed to growth or defense; allocation to one role is thought to occur at the expense of the other (Bryant et al., 1983; Herms and Mattson, 1992; Koricheva et al., 1998). This tradeoff illustrates a possible competition for resources between primary and secondary metabolism where elevated nitrogen to support compensatory growth for example, would result in reduced PSM for chemical defense (Bryant et al., 1987; Muzika and Pregitzer, 1992).

Fenced areas built to exclude browsing in locations having high ungulate density have proven to be effective in allowing the growth and recovery of balsam fir and more palatable tree species (McInnes et al., 1992; Casabon and Pothier, 2008; Smith et al., 2010). An early study of forest regeneration in Newfoundland showed that moose exclosures led to four times as many live balsam fir stems within exclosures than in unprotected areas and that within exclosures, mean annual height growth of protected saplings was more than three times greater than for those outside of exclosures (Thompson et al., 1992). Exclosures have been used to study the impact of moose on fir-dominated boreal forests from perspectives such as species composition, natural regeneration, competition and ecosystem processes (McInnes et al., 1992; Thompson et al., 1992; Pastor et al, 1993; McLaren et al., 2009; Ellis and Leroux, 2017). Few studies; however, have used exclosures in North America to examine the response of chemical defense systems of trees released from intense browsing pressure. In African savannas, total phenol and condensed tannin levels in exposed woody plants were generally lower with high levels of browsing levels when compared to those protected by herbivore exclosures; however, plant chemical responses to browsing were species-specific (Scogings et al., 2013; Wigley et al., 2015).

Throughout GMNP, nine trial moose exclosures were erected between 1995 and 2009 thereby providing a unique opportunity to compare the chemistry of balsam fir foliage under extremes of browsing pressure and possibly gain insight to why moose consume so much balsam fir here. A better understanding of the chemical ecology of browsed and unbrowsed fir saplings could provide insights into strategies employed by balsam fir to persist under heavy browsing pressure. Given that moose are important drivers of community structure and succession of balsam fir forests (Connor et al., 2000; Smith et al., 2010), such knowledge could inform management approaches to mitigate forest decline (Wooley et al., 2008). Our first objective was to examine growth-related responses of balsam fir saplings to the removal of moose browsing pressure using indicators of sapling growth and health (e.g. height-diameter relationships, browsing damage). Secondly, we wanted to observe the effect browsing removal had on levels of foliar chemicals indicative of resource allocation to: i) primary metabolism/growth/high forage quality (e.g. carbon, nitrogen) and ii) secondary metabolism/herbivory defense/low forage quality (e.g. total phenols and condensed tannins). Finally, we intended to explore relationships among foliar chemical traits and to relate these to sapling (e.g. height, diameter, browsing status) and site (canopy openness) factors. Comparisons of exposed and protected saplings should allow us to observe whether balsam fir emphasizes tolerance or resistance (constitutive chemical defenses) as a strategy to cope with intense browsing. We hypothesize that release from browsing pressure will relax constitutive levels of costly (Gershenzon, 1994; Bryant and Julkunen-Tiitto, 1995) defense which could free resources to support greater vertical growth in protected saplings.

2. Methods

2.1. Study area and site location

Field studies were conducted from July 2 to 17, 2012 in Gros Morne National Park (GMNP) on the west coast of Newfoundland, Canada (6400302 N, 578,028 E). Established in 1973, the park covers an area of 1805 km^2; 43% of which is forest dominated by balsam fir with black spruce and white spruce also having importance (Rose and Hermanutz, 2004). The region has a cool, moist climate due to its coastal location on the Gulf of the St. Lawrence (Thompson et al., 2003). The mean annual air temperature (Rocky Harbour) is 3.6 °C, with a mean annual daily minimum and maximum temperature of −0.4 °C and 7.5 °C, respectively (Environment Canada, 2013). The area has a mean annual rainfall of 898 mm and an average annual snowfall of 418 mm.

Studies were conducted at eight moose exclosures located at: Baker's Brook (BB); Killdevil (KD); Martin's Point (MP); Millbrook (MB); Norris Point (NP); Saint Paul's (SP); South East Brook (SB) and Western Brook Pond (WB) (Table 1). Most exclosures were installed in 1995 with exceptions being BB (2009), WB (2008) and KD (2005). All exclosures were 15 × 15 m in size with the exception of BB (100 × 100 m).

2.2. Sapling measurements and foliage sampling

We planned to sample only saplings (height > 0.5 m,

Table 1
Location and description of the eight moose exclosures in GMNP used in this study.

Exclosure (Code)	Year Established	Years of Release	Dimensions (m)	UTM Coordinates	Moose density[a] (moose km^{-2})
Baker's Brook (BB)	2009	3	100 × 100	434,016 E 5498603 N	1.88
Killdevil (KD)	2005	7	15 × 15	445,413 E 5477902 N	1.50
Martin's Point (MP)	1995	17	15 × 15	435,974 E 5514192 N	1.13
Millbrook (MB)	1995	17	15 × 15	443,471 E 5483763 N	6.68
Norris Point (NP)	1995	17	15 × 15	437,427 E 5490502 N	0.88
Southeast Brook (SB)	1995	17	15 × 15	451,157 E 5479124 N	5.25
St. Paul's (SP)	1995	17	15 × 15	449,142 E 5521513 N	6.81
Western Brook Pond (WB)	2008	4	15 × 15	438,921 E 5514863 N	0.75

[a]Estimates from a 2009 moose census (Taylor and Knight, 2009) for sample blocks in close proximity to moose exclosures. Moose density for Millbrook is from an unpublished 2015 Parks Canada survey.

DBH < 10 cm); however, at two exclosures (SB and SP) saplings were limited in number and some balsam fir trees (> 10 cm DBH) were used to achieve balanced replication. Despite having a large DBH, most sampled trees were relatively low in stature and all had lower branches that would easily be reached by moose. Hereafter, all sampled balsam fir is referred to as saplings. At each of the eight moose exclosures, 12 protected (inside exclosures) and 12 exposed (outside exclosures) saplings were randomly selected and tagged for a total of 192 saplings in the study. Exposed saplings were at least four meters away from the exclosure fence to minimize the influence of the physical structure on browsing behavior (McLaren et al., 2004).

The height, basal diameter and related canopy closure was measured for each tagged balsam fir sapling. Percent canopy closure was estimated using a convex spherical densitometer (Model A, Forestry Suppliers, Jackson, MS, USA) (Lemmon, 1956) and was based on the average of four readings taken at each of the cardinal directions at a height of 2.0 m and a distance of 0.5 m from the sapling drip edge. Each sapling was examined for the degree of apical (removal of leader) and lateral browsing. In cases where saplings were apically browsed, regrowth was recorded if the sapling was re-establishing an apical shoot. Three intensities of lateral browsing were noted; no browsing; light browsing (< 30% of lateral branches browsed; still having much of the previous year's needles present) and intense browsing (> 30% of lateral branches browsed or having little to no foliage from the previous year).

For each measured balsam fir, current-year foliage was sampled by removing terminal portions of lateral branches at each of four cardinal directions and two heights representing ⅓ and ⅔ of the sapling height. A subsample of foliage intended for analysis of total phenols (hereafter phenols) and condensed tannins (hereafter tannins) was frozen immediately to prevent chemical degradation. The remainder of the foliage sample, intended for C and N analysis, was collected and stored in a paper bag.

2.3. Sample preparation and chemical analyses

Foliage intended for C and N analysis was dried for 48 h in an oven at 50° C. The needles were removed from the stems and finely ground. C and N analyses were conducted at the Forest Resource and Soil Testing Laboratory, Lakehead University, Thunder Bay, Ontario, using a vario EL cube (Elementar). Whole frozen needles were used for phenol analysis. Phenols were analyzed using a modified Folin-Denis assay (Waterman and Mole, 1994) and extracted from frozen foliage with 40% (v/v) ethanol via sonication for 30 min. Fifty µL of the extracted supernatant was pipetted off and placed into a glass cuvette to which 3.95 mL of deionized water was added along with 250 µL of 1 N Folin-Ciocalteu's Phenol reagent and 750 µL of sodium carbonate (20% w/v). After processing the extract, the phenolic content was measured via spectrophotometry (Thermo GENESYS 20) and phenol concentration was calculated using Beers Law and a gallic acid standard.

Frozen foliage intended for condensed tannin analysis was ground into a fine powder, using dry ice and a mortar and pestle, immediately before analysis. Tannins were analyzed as outlined by Makkar (2003) from frozen foliage via n-butanol-HCl hydrolysis. For each sample the freshly ground balsam fir was first extracted in 10 mL of 70% (v/v) acetone via sonication for 20 min and then centrifuged for 10 min at 3000 rpm at 4 °C. The extracted supernatant was pipetted into a test tube so the extraction could be done on the same tissue again, with 5 mL of 70% (v/v) acetone and the same sonication and centrifuge conditions as the first extraction. Upon completion of the re-extraction, the supernatant was added to the first extracted supernatant. The samples were covered and stored at 3 °C prior to analysis. For tannin analysis, 500 µL of the extracted supernatant was pipetted off into a test tube and diluted to 1:5 with 70% (v/v) acetone. To each test tube, 3 mL of n-butanol-HCl (95:5 v/v) was added along with 100 µL of Ferric Reagent (2% in 2 N HCl). The sample was then placed in a sand bath set at 85–90 °C for 1.5 hr. After processing the extract, the tannin content was measured via spectrophotometry (Thermo GENESYS 20) and tannin concentration was calculated using Beers Law and a tannic acid standard.

2.4. Statistical analysis

Moose exclosures were purposely installed by Parks Canada to span different stand ages and moose densities. Because these and other factors such as exclosure age, canopy closure, tree height and time since last insect disturbance differed among exclosure locations, we made pair-wise comparisons of growth and chemistry variables for protected and exposed saplings for each exclosure, rather than solely relying on the analysis of pooled data for all exclosures. IBM SPSS Statistics 26 was used for data exploration and analysis. Data for several of the measured variables were not normally distributed and transformation failed to normalize all variables. We therefore analyzed untransformed data using non-parametric tests. The significance of differences in means for measured and derived variables between protected and exposed saplings were determined using a Mann-Whitney *U* test followed by a calculation of the effect size (Eta^2) (Field, 2009). A Spearman's Correlation test on pooled data was used to determine whether significant relationships existed among site/growth and foliage chemistry variables. A Mann-Whitney *U* test was also conducted for saplings outside of moose exclosures to determine whether significant differences existed for foliage chemistry variables for each of the various observed browsing conditions (apical browsing, apical regrowth, lateral browsing and browsing intensity).

Despite that assumptions of normality and homogeneity of variance could not be met for all dependent variables, we used (parametric) analysis of covariance (ANCOVA) to examine the overall effect of EXCLOSURE (location) and POSITION (inside or outside the exclosure) on needle chemistry data pooled for all exclosures. Canopy closure and sapling height were used as covariates in this analysis. We justified this approach because hypothesis testing using ANOVA/ANCOVA models is robust to violations of underlying assumptions if violations are not severe, if calculated p values are not close to α, and if no alternative non-

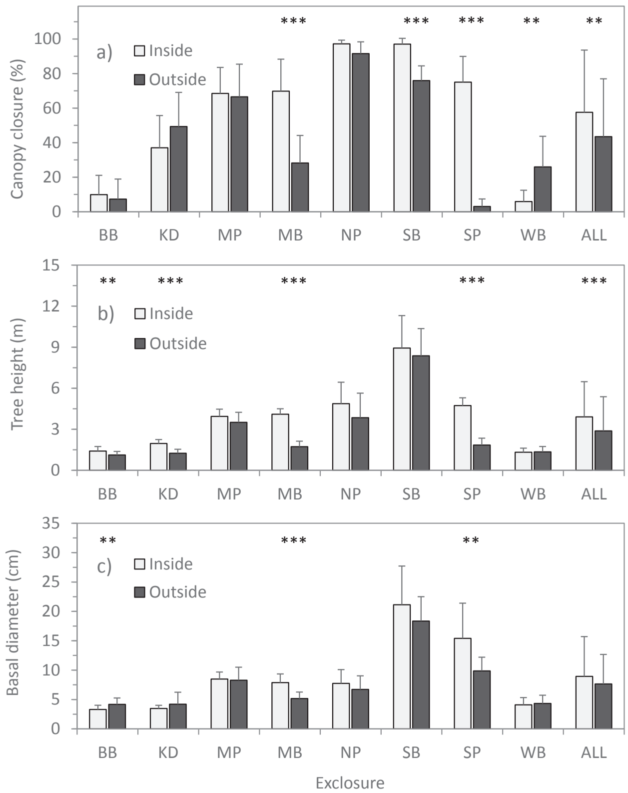


Fig. 1. Mean (+SD) for a) canopy closure, b) height and c) basal diameter related to balsam fir saplings/trees inside and outside of individual (n = 12) and combined for all (n = 96) moose exclosures in GMNP. Where inside and outside values at a particular location statistically differ according to a Mann Whitney *U* test, the level of statistical significance is noted by * = p < 0.05, ** = p < 0.01, or *** = p < 0.001. Differences having no such notation are not significant.

parametric models are available for the experimental design (Zar, 2010). For all tests, statistical significance was considered at p < 0.05.

3. Results

3.1. Site, sapling growth and browsing characteristics

In comparing site and sapling variables inside and outside of moose exclosures, clear and consistent trends were not evident. Outside of exclosures, canopy closure differed greatly among locations ranging from 3.1% at SP to 91.6% at NP. Canopy closure was similar inside and outside of exclosures at BB, KD, MP, and NP. Compared to outside of exclosures, inside canopy closure was greater at MB, SB, and SP; however, at WB, canopy closure was greater outside the exclosure (Fig. 1). The most significant impact of release from moose browsing was observed at SP where canopy closure outside the exclosure was 3.1% compared with 75% closure inside the exclosure. Height of sampled balsam fir saplings was greater inside exclosures in almost all cases; this trend being less consistent for basal diameter. The average basal diameter of protected and exposed fir saplings was significantly different at only three exclosures. At one of these (BB), exposed saplings had larger diameters than those protected within the exclosure (Fig. 1). Pooling site/sapling data for all exclosures showed that overall, canopy closure and sapling height were significantly greater inside than outside of exclosures; however, the size of the effect was small, accounting for only 4.4% and 7.3%, respectively of the variation due to exclosures relative to location (Fig. 1).

Outside of exclosures, there was a high frequency (83% − 100%) of apical browsing at BB, KD, MB, SP and WB; while outside BB, KD and MB, there was also a high frequency (58% − 100%) of recovery of the apical shoot (leader) (Table S1). Only at MP and NP did we find any fir saplings outside of exclosures that showed no evidence of lateral browsing; in both cases, the frequency of this was low (8.3% − 16.7%). Despite this, instances of light or heavy lateral browsing for unprotected saplings at all locations were high. At MB and SP, all sampled saplings showed heavy lateral browsing; all of these saplings had also been apically browsed (Table S1). Height-diameter relationships illustrate the degree to which growth of balsam fir saplings was suppressed by moose browsing. To attain a common height of 2.0 m, for example, tagged saplings outside of exclosures had to grow radially, on average, to a diameter of more than two times greater than those released from

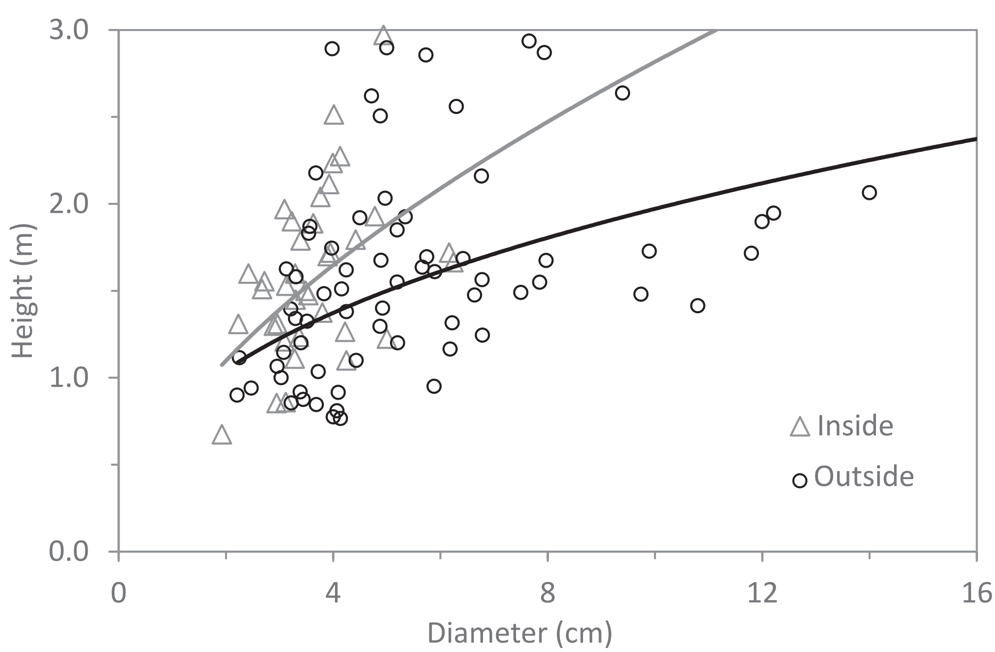


Fig. 2. The relationship between basal diameter and height of balsam fir saplings based on pooled data inside (R^2 = 0.231; n = 37) and outside (R^2 = 0.216; n = 70) of eight moose exclosures. Trend lines and R^2 values are based on power regression models. Saplings in excess of 3.0 m in height (beyond the reach of moose) were excluded from this analysis.

Table 2

Mean (± SD) concentration of carbon (C), nitrogen (N), total phenols (TP) and condensed tannins (CT) for current-year balsam fir foliage from inside and outside of individual (n = 12) moose exclosures and pooled for all exclosures (n = 96) in GMNP. The Z, *p* (asymptotic, 2-tailed) and Eta^2 values are derived from a Mann Whitney *U* test (n = 12) and are bolded where "inside" and "outside" values differ significantly (p < 0.05).

Exclosure	Position	C (% DW)	N (% DW)	C/N	TP (mg g^{-1} FW)	CT (mg g^{-1} FW)
Baker's Brook	Inside	53.89 ± 0.40	2.09 ± 0.46	27.22 ± 7.21	8.29 ± 3.21	1.48 ± 0.49
	Outside	54.05 ± 0.36	2.03 ± 0.17	26.79 ± 2.25	11.76 ± 1.84	1.23 ± 0.45
	p	0.299	0.488	0.603	**0.010**	0.204
Killdevil	Inside	56.07 ± 6.71	1.65 ± 0.19	34.18 ± 3.74	7.68 ± 2.70	0.94 ± 0.52
	Outside	53.96 ± 0.36	1.58 ± 0.17	34.61 ± 3.72	9.68 ± 2.50	0.91 ± 0.43
	p	0.248	0.356	0.729	0.083	0.773
Martin's Point	Inside	53.54 ± 0.40	1.23 ± 0.13	43.84 ± 4.51	7.27 ± 2.63	0.71 ± 0.35
	Outside	53.68 ± 0.36	1.31 ± 0.24	42.16 ± 6.19	5.23 ± 1.17	0.68 ± 0.15
	p	0.326	0.386	0.419	**0.021**	0.686
Millbrook	Inside	53.56 ± 0.48	1.78 ± 0.28	30.79 ± 4.34	4.75 ± 1.55	0.73 ± 0.45
	Outside	54.20 ± 0.75	1.49 ± 0.22	37.01 ± 5.21	9.12 ± 1.85	1.19 ± 0.37
	p	**0.021**	**0.013**	**0.008**	**< 0.001**	**0.015**
Norris Point	Inside	53.13 ± 0.51	1.73 ± 0.13	30.94 ± 2.36	6.44 ± 1.73	1.30 ± 0.36
	Outside	53.66 ± 0.59	1.77 ± 0.28	30.96 ± 4.16	3.78 ± 1.16	1.23 ± 0.55
	p	**0.024**	0.525	0.488	**0.001**	0.371
Southeast Brook	Inside	52.82 ± 0.49	1.71 ± 0.22	31.42 ± 3.88	1.89 ± 1.37	1.05 ± 0.64
	Outside	53.41 ± 0.54	1.98 ± 0.24	27.36 ± 3.21	5.23 ± 1.94	1.25 ± 0.41
	p	**0.009**	**0.013**	0.018	**< 0.001**	0.119
St. Paul's	Inside	54.24 ± 0.52	2.54 ± 0.49	22.01 ± 3.96	2.90 ± 1.47	1.07 ± 0.31
	Outside	54.15 ± 0.48	1.84 ± 0.28	30.06 ± 4.14	8.97 ± 1.65	1.23 ± 0.45
	p	0.817	**< 0.001**	**< 0.001**	**< 0.001**	0.686
Western Brook Pond	Inside	54.09 ± 0.53	1.57 ± 0.13	34.73 ± 2.63	5.57 ± 1.56	0.76 ± 0.20
	Outside	54.12 ± 0.70	1.48 ± 0.15	36.88 ± 3.58	10.07 ± 1.64	1.06 ± 0.42
	p	0.419	0.166	0.248	**< 0.001**	**0.035**
All	Inside	53.56 ± 0.48	1.79 ± 0.46	31.89 ± 7.23	5.60 ± 2.98	1.01 ± 0.49
	Outside	53.90 ± 0.58	1.68 ± 0.33	33.23 ± 6.46	7.98 ± 3.17	1.10 ± 0.45
	Z	−2.309	−1.042	−1.094	−4.931	−1.696
	p	**0.021**	0.298	0.274	**< 0.001**	0.090
	Eta^2	0.028	0.006	0.006	0.127	0.015

browsing (Fig. 2).

3.2. Foliage chemistry

Foliar C levels tended to be greater in saplings outside of exclosures; however, differences were only significant at MB, NP, SB and in the case of data pooled for all exclosures (Table 2). Foliar N levels were similar for inside and outside saplings at most exclosures (BB, KD, MP, NP and WB). At MB and SP, N levels were greater for outside saplings than those inside; however, the opposite trend was observed at SB (Table 2). Inside and outside saplings had similar C\N ratios at most exclosures with the exception of MB and SP, where outside saplings had significantly greater C\N ratios, and SB, where this ratio was greater for inside fir saplings (Table 2).

Foliar phenol concentration varied considerably; saplings released from browsing pressure had mean phenol levels as low as 1.89 mg g^{-1} (SB) compared to outside mean values that reached a high of 11.76 mg g^{-1} (BB). Foliar phenol levels were most responsive to release from browsing and consistently differed between inside and outside locations. In most cases, phenol levels for outside saplings were greater than those inside exclosures, with the exception of MP and NP, where phenol levels were significantly greater for inside saplings (Table 2). Within the oldest exclosures (MB, MP, NP, SB and SP; 17 years of moose browsing release), foliar phenol levels averaged 5.6 ± 2.83 mg g^{-1} compared to 8.8 ± 2.99 mg g^{-1} inside of the newest exclosures (BB, KD, and WB; 3 – 7 years of release). Trends for tannin did not show the extent of inside-outside differences observed for TP; however, balsam fir saplings at MB and WB did have significantly greater concentrations of foliar tannins outside of exclosures than those inside. Compared to TP, tannin levels were far less variable among exclosures; ranging from a mean of 0.68 mg g^{-1} (MP) to a high of 1.48 mg g^{-1} (BB) (Table 2). Comparing pooled data in non-parametric tests for all exclosures showed that among foliage chemistry variables, only C and phenol levels differed significantly between saplings located inside and outside of exclosures. For both C and phenols, levels were greater inside exclosures; however, effect sizes were small, accounting for only 2.8% and 12.7% respectively, of the variation (Table 2). Controlling for the covariates canopy closure and sapling height in ANCOVA, revealed that N, tannins and phenols differed significantly between inside and outside exclosure positions (Table S2) when only C and phenols did so according to non-parametric analysis of data pooled for all exclosures (Table S2). With the exception of C, chemistry variables for fir needles differed among exclosures and were affected by the interaction of position × exclosure (Table S2).

An analysis of pooled data for inside and outside saplings showed that levels of both foliar C and phenols were negatively correlated with the height and basal diameter of saplings, and canopy closure (Table 3). For any given height or degree of canopy closure, phenol levels were consistently greater for saplings outside of exclosures (Fig. 3). Phenol levels were positively related to those of C, while tannins were positively related to N levels and negatively related to the C\N ratio (Table 3). N was strongly related to the C\N ratio; however, surprisingly, there was no significant relationship between foliar C and the C\N ratio (Table 3).

Outside of moose exclosures, apically browsed fir saplings showed higher foliar C levels than unbrowsed saplings (Table S3). Similarly, C concentration was also greater in laterally browsed saplings. Foliar phenol levels were greater in saplings that: 1) were heavily browsed more so than those lightly browsed; 2) had undergone apical browsing; and 3) were attempting to restore an apically dominant leader. Foliar N, C\N ratio and tannin levels were consistently unresponsive to browsing pattern and intensity (Table S3).

4. Discussion

4.1. Sapling height and herbivory defense

In GMNP, intense browsing, especially of leaders, suppresses

Table 3
Relationships among site, growth and foliage chemistry variables indicated by Spearman's correlation coefficients (rho) (n = 192, pooled data for inside and outside of exclosures) and associated *p* values which are bolded for cases where relationships are statistically significant ($p < 0.05$). See Table 2 for chemical abbreviations.

		Height	Diameter	C	N	TP	CT
Canopy closure	rho	0.766	0.538	−0.473	−0.004	−0.568	0.010
	P	< **0.001**	< **0.001**	< **0.001**	0.952	< **0.001**	0.890
Height	rho		0.821	−0.417	0.103	−0.646	−0.008
	p		< **0.001**	< **0.001**	0.154	< **0.001**	0.909
Diameter	rho			−0.284	0.156	−0.501	−0.011
	p			< **0.001**	0.031	< **0.001**	0.878
C	rho				0.127	0.280	−0.022
	p				0.078	< **0.001**	0.760
N	rho					−0.110	0. 272
	p					0.130	< **0.001**
TP	rho						0.126
	p						0.081

vertical growth and keeps saplings within the reach of moose where they remain vulnerable to rebrowsing and where the chance to re-establish apical dominance and recruitment to the canopy is reduced. Inside exclosures, release from browsing pressure allows significant height growth in protected saplings as also found in other exclosure studies (Risenhoover and Maass, 1987; Thompson et al., 1992). At sites where unprotected saplings and young trees have grown vertically beyond the reach of moose (e.g. SB), apical damage and lateral browsing is reduced despite saplings having accessible branches and a relatively high moose density in the vicinity. An exception to these trends was reported by McLaren et al. (2009) who found that within exclosures, the removal of browsing suppression of balsam fir growth was replaced by suppression from competition with fast-growing broadleaved tree species.

Release from vulnerability to moose browsing appears to trigger a decline in constitutive levels of phenols, likely arising from a shift in the carbon-nutrient balance required to support more active growth of larger canopies. Taller fir saplings have lower phenol and C concentrations but height is unrelated to N and tannins. The negative relationship between height and phenols suggests a resource tradeoff as energy allocated to growth, indicated in our study by taller saplings having regrown leaders and lateral foliage, likely comes at the expense of production of PSM (Loehle, 1987). This could explain why phenol levels outside MP and NP, where saplings were relatively tall, did not follow the trend of other exclosures. Having reached a critical height and become less vulnerable to moose browsing, saplings outside of these exclosures appear to have reduced constitutive phenol levels allowing resource investment in vertical growth. This is likely facilitated by the relatively low moose densities at these exclosures. That escaping moose browsing requires prioritized resource allocation to support rapid vertical growth (Brandner et al., 1990) exemplifies the "grow or defend dilemma" of plants (Herms and Mattson, 1992) as such a resource trade-off would increase the palatability of poorly defended tissues. Under high browsing pressure, allocation to defense at the expense of growth would prolong the time over which saplings remain entirely within reach of moose. Such a situation where browsers keep saplings in a constant state of accessibility through a positive feedback loop (browsing–regrowth–rebrowsing) has been called a browsing lawn (Fornara and Toit, 2007), a modification of the herbaceous grazing lawn concept described by McNaughton (1984). Especially under high browsing pressure, height is an important factor in the chemical and regeneration ecology of balsam fir.

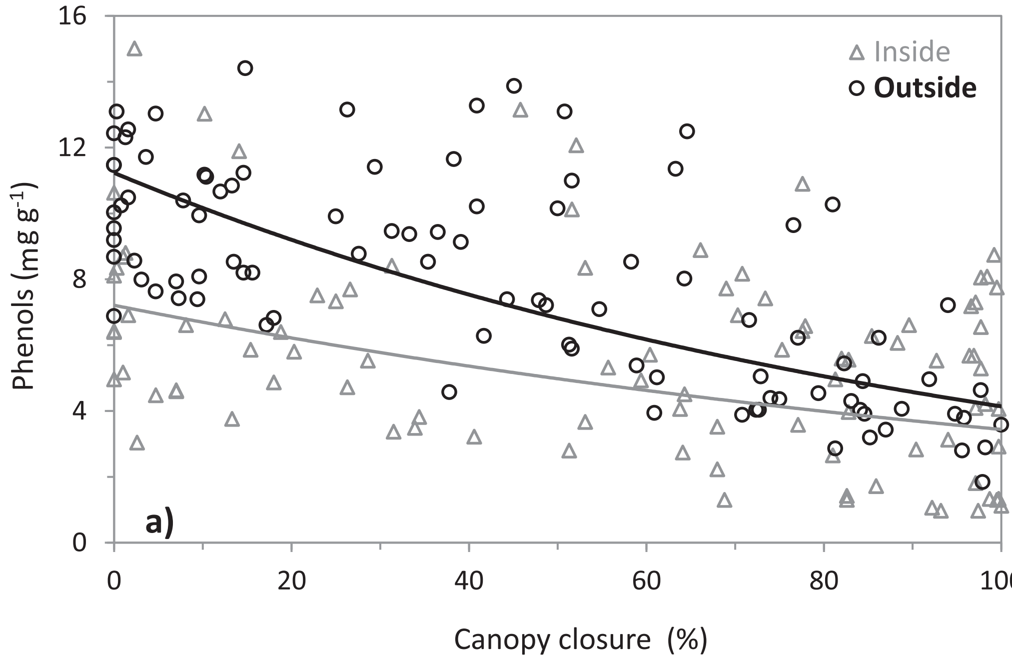
4.2. Foliar chemistry

Phenol levels were significantly greater for exposed compared to protected saplings; apically or laterally browsed saplings compared to those unbrowsed; and heavily browsed compared to lightly browsed saplings. This is in keeping with the classic understanding of how browsing pressure leads to the enhanced production (induction) (Schowalter et al., 1986; Karban and Myers, 1989; Bryant et al., 1992; Stolter et al., 2005) or higher constitutive levels (Vourc'h et al., 2001) of PSM as a means of herbivory defense. Our study design does not permit conclusions regarding induction of chemical defenses following browsing; however, using protected saplings as a reference suggests that moose browsing pressure over the longer term elevates the constitutive levels of phenols in balsam fir.

Differences among exclosures in phenol concentration are likely due to differences in moose density and browsing pressure, as well as stand characteristics that affect light interception and availability of C. The exclosures at MP and NP, where phenol levels were significantly greater inside than outside of exclosures, were of an older vintage, situated in more mature and closed forest and in areas of relatively low moose density and browsing disturbance. Here, canopy closure, sapling height, and basal diameter; traits that were negatively correlated with foliar phenol levels, were relatively high. In balsam fir foliage, phenol levels were positively correlated with levels of C but were unrelated to tissue N. This is consistent with the functioning of phenols as relatively low cost C-based PSM that are more likely to be limited by the availability of light than N (Dudt and Shure, 1994; Koricheva et al., 1998).

Unlike phenols, tannins were generally unaffected by differences in light availability (canopy closure), sapling growth characteristics or browsing pressure. At most exclosures, tannin levels were similar for exposed and protected saplings implying no adjustment in levels of this PSM group to browsing pressure. Based on a nonparametric test of pooled data for all locations, foliage tannin levels outside and inside exclosures did not significantly differ. However, controlling for canopy closure and sapling height using parametric testing, suggested that foliage tannins levels were significantly larger outside of exclosures. This finding must be considered cautiously; the actual mean difference in tannins was very small and the underlying assumptions of the parametric test could not be met for tannin data. Furthermore, tannin levels were unrelated to canopy closure and sapling height based on a nonparametric correlation analysis. It is unclear why tannin levels were significantly greater for exposed than protected saplings at only the MB and WB exclosures. These exclosures had a relatively high moose density and/or a high frequency of intensely browsed saplings suggesting that costlier tannins might only be activated once a critical threshold of biomass removal was surpassed.

Studies of coniferous tree species have consistently reported that phenol levels are high in spring and decline over the growing season, especially in current-year foliage (Wagner et al., 1990; Nerg et al., 1994; Hatcher, 1990; Zou and Cates, 1995; Grégoire et al., 2014; Nosko and Embury, 2018). Our foliage samples were collected early in the growing season shortly after bud-burst; a time when tissue phenol levels would be relatively high. This is likely a significant factor influencing the responsiveness of phenols to differences in light availability, tree height and browsing pressure. The lack of elevated tannin levels in



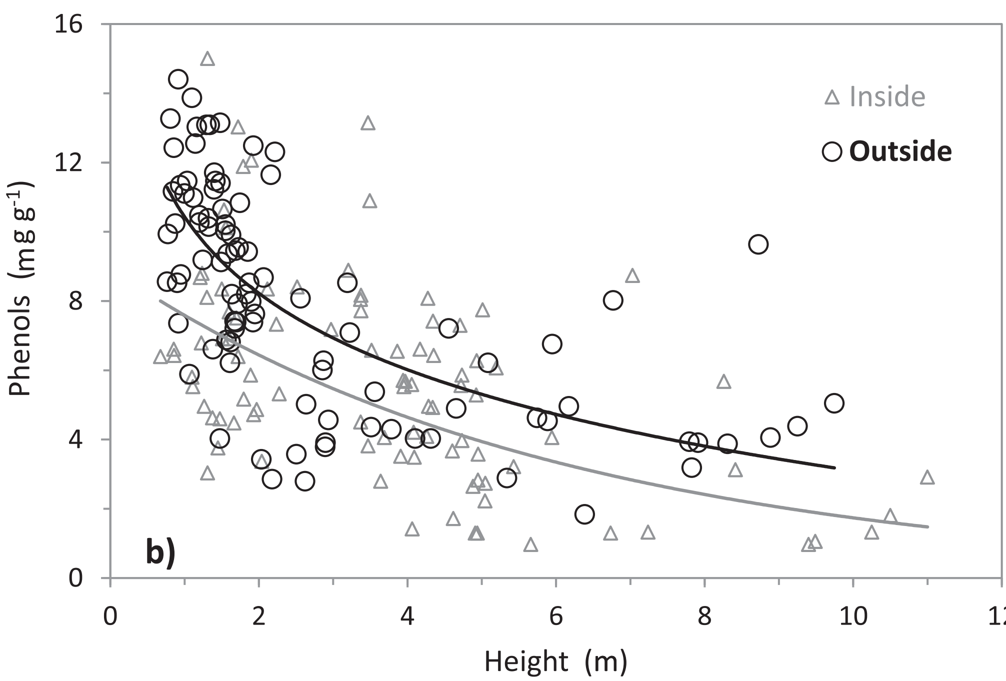


Fig. 3. The relationship between a) canopy closure (R^2 inside = 0.170, R^2 outside = 0.536, based on exponential regression models) and b) sapling height with foliar phenol concentration using pooled data (R^2 inside = 0.367, based on an exponential regression model; R^2 outside = 0.473, based on a logarithmic regression model) of eight moose exclosures (n = 96). Trend lines and R^2 values are based on linear regression models.

browsed versus unbrowsed saplings could also be due to the time of year that tissues were sampled. While seasonal patterns for tannins can differ among tree species (Zou and Cates, 1995), many studies of deciduous and coniferous woody plants (Feeny, 1970; Wagner et al., 1990; Zou and Cates, 1995; Riipi et al., 2002; Barbehenn and Constabel, 2011), including balsam fir seedlings (Nosko and Embury, 2018), indicate that in the early growing season, foliar tannin concentrations are low and steadily increase, reaching a peak late in the late growing season.

Because phenols were negatively related to canopy closure, the positive relationship observed between leaf C concentration and phenol level would be expected. Tannin levels did not appear to respond to changes in canopy closure and were not related to foliar C concentration. Reported trends vary; for example, phenol levels in *Larix gmelinii* Rupr. were found to decrease with increasing light availability while tannins increased (Yan et al., 2014). A meta-analysis of 107 deciduous and coniferous species revealed however, that under low light, C-based PSM tend to be reduced in woody plants (Koricheva et al., 1998). Release from browsing did not affect foliar N levels; N was unrelated to canopy or tree variables. Significant differences in N between exposed and protected saplings were observed for three exclosures; however, trends were inconsistent. The two exclosures where foliar N was significantly greater for protected saplings (MB and SP) also had large differences in light availability (canopy closure), sapling height and diameter between the inside and outside of exclosures, as well as having high moose density and severe browsing. Under such conditions many unprotected saplings had little foliage that was older than the current year. Foliage of heavily browsed trees often has high N levels (Fornara and Toit, 2007).

Apical browsing can lead to accelerated growth of remaining shoots (Lehtilä et al., 2000); however, compensatory growth needed to produce new photosynthetic tissue and reestablish apical dominance could deplete reserves of nutrients such as N (Edenius et al., 1993; Villar-Salvador et al., 2015). Our finding that foliage levels of N and tannins are positively related in balsam fir is inconsistent with the premise that primary and secondary metabolism compete for resources and that elevated N results in reduced PSM (Bryant et al., 1987; Muzika and Pregitzer, 1992). It is also inconsistent with often-described growth-defense trade-offs (Bryant et al., 1983; Herms and Mattson, 1992; Koricheva et al., 1998). An examination of soil chemistry could clarify this issue by confirming whether sites having high soil N also had low foliar PSM levels.

As with tannins, the C\N ratio for balsam fir foliage was insensitive to differences in browsing pressure. The mean C\N ratio was significantly greater outside of MB and SP than inside but greater inside SB than outside. The high C\N ratio levels outside of MB and SP are likely indicative of the high moose density, intense browsing of balsam fir and low canopy closure at these sites while the greater concentration of foliar N inside of these exclosures suggests that balsam fir saplings released from browsing do not have to invest in C-based defenses to the same degree as balsam fir under extreme browsing. Working in GMNP, Hamelin (2011) found that foliar C/N ratios were greater in balsam fir saplings located in more recently disturbed areas and in forested areas where regeneration appeared to be successful rather than impaired. An examination of response to defoliation showed no growth-defense trade-off in C allocation of balsam fir (Deslauriers et al., 2015).

4.3. Considerations for future study

Only certain compounds within a group of PSM are likely deter a specific herbivore (Hagerman and Robbins, 1993; Ayres et al., 1997; Stolter et al., 2005; Salminen and Karonen, 2011); however, the importance of a particular secondary metabolite could vary depending on the season (Spalinger et al., 2010; Barbehenn and Constabel, 2011). Future studies should attempt to isolate and identify individual phenols and tannins or other chemical defense compounds (e.g. terpenes) that might play a significant role in the chemical defense of balsam fir against moose and determine how the importance of these compounds might change in going from early-season defense against insects to late-season defense against browsing mammals (Stolter et al., 2005; Salminen and Karonen, 2011). Tannins could be a more significant factor in the chemical defense of balsam fir than our early-season samples would suggest. A clear indication of the role of tannins in the chemical defense of balsam fir necessitates an examination of foliar chemistry in the late growing season when peak levels have likely been reached and moose transition to a winter diet dominated by balsam fir (Zou and Cates, 1995; Riipi et al., 2002; Barbehenn and Constabel, 2011; Stolter et al., 2013).

Few studies have compared the nutritional and defense chemistry of balsam fir to less palatable conifers such as spruce. Sauvé and Côté (2007) reported the concentration of tannins to be greater in white spruce than balsam fir twigs and explained that this could account for the selection by white-tailed deer (*Odocoileus virginianus* Zimmermann) of balsam fir over spruce. In contrast, Fuentealba and Bauce (2016) found that phenols, tannins and all tested monoterpenes were significantly greater in balsam fir than in spruce, again calling into question why fir twigs are so much more palatable to moose and white-tailed deer than those of spruce. A possible reason is that fir in the Fuentealba and Bauce (2016) study also had greater levels of foliar N than white spruce. In Newfoundland, moose select balsam fir twigs having relatively high protein (N) levels despite that concentration of terpenes in these twigs are also relatively high (Thompson et al., 1989). To better understand why balsam fir dominates the winter diet of moose in Newfoundland, the chemistry of balsam fir twigs should be compared to that of other woody forage species and the possibility that defense chemicals are more inducible or present at higher levels in these other forages requires confirmation.

Finally, trends in foliar N, expressed as total N in our case, varied among exclosures but in most cases did not differ between inside and outside saplings. Wigley et al., (2019) found that interpretations of browse quality varied depending on whether total N or available N (DeGabriel et al., 2008) was considered. Based on total N, three species of *Acacia* had similarly high levels of foliage quality; however, when based on available N, foliage quality was clearly higher in one of these *Acacia* species than in the other two (Wigley et al., 2019). Assessment of forage quality based on total N should therefore be used with caution.

4.4. Conclusions

Access to exclosures allowed us to examine the impact of up to 17 years of release from moose browsing on balsam fir. Clearly, high moose density is a significant factor impeding the natural regeneration of fir-dominated forests. Elimination of browsing pressure facilitates forest regeneration as indicated by increased canopy closure, taller saplings having greater basal diameters, and reduced early-season phenol levels suggesting relaxed constitutive defenses and presumably more resources available to support growth. Our data support the hypothesis that removal of browsing pressure lowers constitutive levels of defense chemicals, permitting growth recovery and improved sapling health; however, due to the limitations of our static study design, this supposition is based on indicators rather than direct measures of growth or growth/defense tradeoffs. Our data also suggest that investment in chemical defense by balsam fir appears to be ineffective in deterring moose browsing and the role of induced defenses in balsam fir saplings remains unclear.

Even when exposed to browsing, saplings that can gain sufficient height to escape moose and reestablish apical dominance invest fewer resources in chemical defense. Tissue levels of PSM, at least in the case of phenols, depend on light availability. At high ungulate density, the degree to which balsam fir can persist in the regeneration layer and recruit to the canopy following disturbance could be influenced by the extent to which any residual canopy remains after disturbance. Indications are that balsam fir is adapted to withstand high levels of repeated browsing and allocates significant resources to regrow foliage and reestablish apical dominance if the leader is clipped. In acting as refugia to protect palatable tree species from chronic browsing pressure, exclosures could also provide sources of local genetic stocks of seed for forest restoration programs.

CRediT authorship contribution statement

Peter Nosko: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Supervision. **Kathleen Roberts:** Methodology, Formal analysis, Investigation, Writing - original draft. **Tom Knight:** Conceptualization, Resources, Supervision, Funding acquisition. **Ashley Marcellus:** Methodology, Validation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foreco.2019.117839.

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